

CHICAGO O'HARE, IL, USA

WMO: 725300

Lat: 41.960N

Lon: 87.932W

Elev: 662

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94846

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.7	(c) 3.3	(d) -11.4	(e) 3.1	(f) -0.4	(g) -6.9	(h) 3.9	(i) 4.6	(j) 26.5	(k) 30.5	(l) 24.6	(m) 29.5	(n) 10.6	(o) 270	(p) 0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.6	(c) 91.2	(d) 74.1	(e) 88.5	(f) 72.8	(g) 86.0	(h) 71.5	(i) 77.4	(j) 87.2	(k) 75.5	(l) 84.5	(m) 74.0	(n) 82.5	(o) 11.4	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	131.6	83.4	72.6	124.0	81.3	71.0	117.1	79.3	41.2	87.0	39.4	85.1	37.8	82.7	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	20.8	18.9	-8.2	95.4	6.9	3.6	-13.2	98.1	-17.2	100.2	-21.1	102.2	-26.1	104.9
			-9.0	79.9	6.6	1.8	-13.7	81.2	-17.6	82.2	-21.3	83.2	-26.1	84.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	24.1	27.8	38.1	49.1	59.7	69.8	74.6	73.1	66.0	53.7	40.7	29.7		
	DBStd	19.78	11.77	11.35	10.98	9.11	8.89	7.41	5.85	5.27	7.53	8.81	9.63	10.71		
	HDD50	2961	803	625	397	122	13	0	0	0	1	62	305	633		
	HDD65	6157	1268	1042	835	483	217	37	3	4	78	366	730	1094		
	CDD50	3201	0	3	30	96	314	594	763	715	481	176	26	3		
	CDD65	919	0	0	2	7	53	181	301	254	107	14	0	0		
	CDH74	8096	0	0	19	86	527	1722	2811	1947	871	113	0	0		
	CDH80	2669	0	0	2	14	152	602	1033	589	257	20	0	0		
(14) Wind	WSAvg	9.6	10.8	10.7	10.7	11.1	9.8	8.6	8.1	7.6	8.2	9.5	10.2	10.1		
(15) Precipitation	PrecAvg	35.7	2.0	1.9	2.3	3.3	4.1	4.1	3.8	4.0	2.9	3.0	2.5	1.9		
	PrecMax	46.3	4.6	6.1	5.0	7.3	7.5	10.5	8.1	9.0	10.3	8.4	6.0	4.6		
	PrecMin	22.5	0.5	0.5	0.8	1.3	0.4	1.0	1.7	1.1	0.5	0.9	0.3	0.5		
	PrecStd	5.4	1.1	1.1	1.1	1.4	1.7	2.0	1.6	1.9	2.0	1.9	1.5	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	62.4	76.6	81.9	89.0	93.0	96.5	92.6	90.7	83.0	69.1	60.8		
		MCWB	53.6	54.2	62.9	63.2	69.6	73.7	76.8	75.8	72.8	68.8	57.5	57.1		
	2%	DB	49.4	54.3	67.7	75.8	84.3	89.9	91.6	89.0	86.5	77.0	63.7	54.0		
		MCWB	44.8	48.3	56.6	60.8	67.8	72.5	75.7	73.9	70.8	64.3	56.0	49.9		
	5%	DB	43.2	47.6	60.7	70.8	80.1	86.6	88.4	85.9	82.7	72.5	60.0	48.4		
		MCWB	39.5	42.1	52.6	57.4	65.4	71.2	73.6	72.4	68.8	62.1	53.7	44.6		
	10%	DB	39.1	42.3	54.8	65.4	75.7	83.3	85.8	83.1	79.2	68.2	55.5	43.8		
		MCWB	36.4	38.5	47.6	53.7	63.6	69.8	72.0	70.5	67.0	59.8	49.4	40.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.4	56.2	63.8	66.4	73.5	77.7	80.3	78.8	75.4	71.0	61.6	58.1		
		MCDB	56.3	60.7	75.2	77.6	84.7	87.6	90.6	89.5	84.9	80.2	66.0	60.2		
	2%	WB	46.0	48.8	58.7	62.1	70.3	75.2	78.1	76.4	73.2	67.2	58.1	50.8		
		MCDB	48.3	54.6	65.5	72.3	80.0	85.7	87.9	85.1	82.1	72.9	62.5	53.0		
	5%	WB	40.2	42.7	53.8	59.0	67.9	73.2	76.4	74.7	71.2	64.0	53.8	44.8		
		MCDB	42.6	46.9	59.9	69.3	77.0	83.3	85.7	82.4	79.2	70.2	58.8	47.8		
	10%	WB	36.6	38.6	47.7	55.1	65.1	71.4	74.5	73.1	69.2	60.7	50.1	40.3		
		MCDB	39.1	42.1	54.4	64.3	73.5	80.9	83.1	80.0	76.7	67.1	55.0	43.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	14.1	15.8	18.4	19.2	18.9	17.6	16.5	18.0	16.7	14.3	12.3		
		MCDBR	16.8	20.7	23.7	27.3	24.4	21.9	19.8	19.5	21.8	22.9	19.9	16.6		
	5% WB	MCWBR	14.6	15.9	16.0	16.0	12.3	10.0	8.6	8.4	9.4	13.0	14.5	13.9		
		MCWBR	16.0	19.2	22.1	24.5	21.4	19.7	18.3	16.7	18.5	19.0	18.0	15.4		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.314	0.347	0.401	0.428	0.438	0.446	0.440	0.411	0.358	0.316	0.298		
		taud	2.435	2.383	2.382	2.252	2.230	2.249	2.228	2.242	2.291	2.439	2.507	2.470		
	Ebn at Noon		270	283	285	275	270	266	263	260	260	263	262	260		
		Edh at Noon	24	30	34	42	44	43	43	42	37	29	23	22		
(44) All-Sky Solar Radiation	RadAvg		502	781	1134	1466	1676	1919	1933	1685	1392	912	579	418		
	RadStd		57	99	90	120	136	129	95	100	118	101	57	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	-0.92	-1.08	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page